University of Rwanda.
College of Science and Technology
ICT
Computer Engineering
Mobile application

Offline posts manager.

1. Dependencies used and why.

The pubspec.yaml file declares the following key dependencies:

Package	version	Purpose.
sqflite	^2.3.3+1	SQlite database engine for Flutter.
path	^1.9.0	Safely constructs file system paths
path_provider	^2.1.4	Locates platform-specific storage directories
uuid	^4.5.1	Generates unique IDs for each post.
image_picker	^1.1.2.	Allow picking images from the gallery/ camera
provider	(via Flutter)	State management - connects DB layer to UI.
flutter_slidable.	^3.1.1	Swipe-to-delete gesture on post cards.
shimmer	^3.0.0	Loading skeleton placeholders.
intl	^0.20.2	Date formatting
flutter_staggered animation	^1.1.1	Animated list entry effects.

Why sqlite is necessary for local storage.

Its core design requirement is that it must work with no internet connection. This rules out any cloud-based or server-dependent storage solution.

reasons to choose sqlite:

a) serverless and self contained.
b) relational structure with SQL query power.
c) persistence across app restarts.
d) no network dependency.

2. How database Exception are Handled.

> Database not Initialized.

The DatabaseHelper uses the Singleton pattern with lazy initialization to guarantee the database is opened exactly once before any operation.

> Insert/Update/Delete Errors.

Error handling for write operations is delegated to PostsProvider, which wraps every database call in a try/catch block. Each method returns a bool to signal success or failure to UI layer.

> Invalid or corrupted Data

The Post.fromMap() factory constructor is the point where raw SQLite row data is deserialized back into a Dart object.

3. SQLite in Flutter.

SQLite disk I/O is a blocking operation that, if run on the main UI thread, would freeze animations and gestures. Flutter handles this through Dart's async/await concurrency model.

the sqflite package runs SQLite on a dedicated background isolate.

the Future returned by db.query() or db.insert() resolves once the background isolate completes the work.

CamScanner